

cryptographic technique used to verify the authenticity, integrity & non-repudiation of digital message.
(read, unchanged & sent by a correct person)

Digital Signatures (ensures that message was created by a known sender & it has not been altered during transmission.)

1) Key Generation Algorithm $\rightarrow$ Digital Signature uses asymmetric cryptography

This key pair ensures secure authentication during digital transaction.

- Private Key:- used to create signature
- Public Key:- used to verify signature.

2) Signing Algorithm $\rightarrow$ To create a digital signature:-

- Hash function is applied on original message to generate fixed-length value called message digest.
- The message digest is then encrypted using sender's private key.
- The encrypted hash value forms the digital signature.

Only the hash is encrypted.

- Hash values are shorter
- Hashing is faster than encryption
- Improves efficiency without reducing security.

3) Signature Verification Algorithm

At receiver side:-

- Digital signature is decrypted using sender's public key, producing original message digest.
- Receiver independently computes the hash of received message using same hash function.
- Both hash values are compared:

Digital Signature = Encryption (Sender's private key, message digest)

Date:

If they match (Signature valid)
If they differ (Not valid)

Sender → hash → encrypt with private key → digital signature.
Receiver → decrypt with public key → Compare hashes → Verify.

WEB SECURITY

- Why server-side web apps are hard to secure?
- Attackers can not see server code directly, but they can send bad input
- Normal security can not stop attack becz attack looks like normal HTTP request. Also many security devices do not check what is inside HTTP requests.
- Zero day attack → • Attacker use new vulnerability so no time to prepare for it.

ADDRESS RESOLUTION PROTOCOL (ARP)

- Protocol to find MAC Addresses of a device when we already know its IP addresses.
ARP works only inside a LAN, not across network.

- Why ARP needed?

ARP solves this {
• Host A wants to send data to B
• Both on same LAN
• Host A knows B's IP, not B's Mac.